CURRICULUM VITAE



Shijie Xu

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Date of Birth

March 9th, 1995

Education

Ph.D. in Environmental Science, Hokkaido University — March 2025
Dissertation Title: "Accurate and Fast Predictions of Protein Structures and Properties via Protein Language Models"
M.S. in Computer Science, University of Science and Technology of China — June 2020
B.S. in Mathematics, University of Science and Technology of China — June 2017

Research Experience

Postdoctoral Researcher, Hokkaido University — April 2025 - Present

Research Interests

AI for Science, Protein Language Models, Bioinformatics, Computational Chemistry, Deep Learning

Skills

Programming: Python / C++ / CUDA / LaTeX/ FORTRAN / Rust / Mathematica / MATLAB / Lean / Coq / Haskell
Web development: HTML / JavaScript / FastAPI / Flask / Django / Go
Software: AutoDock Vina / PyMOL / Git / Docker / Singularity / VSCode
Hardware: GPU workstation building / Server maintenance / Raspberry Pi
Languages: English (fluent), Japanese (basic), Chinese (native)

Publications

1. Shijie Xu and Akira Onoda*. "Probe-Based Identification of Metal-Binding Sites Using Deep Learning Representations." *Nature Communications* 17: 8146, **2026**, *DOI:* https://doi.org/10.1038/s41467-026-74657-x

2. Shijie Xu, Ke Xu, and Akira Onoda*. "Accurate Prediction of Enzymatic Degradation of Plastics by Language Models." *Chemistry Letters* 55(2): upag022, **2026**, *DOI:* https://doi.org/10.1093/chemle/upag022
3. Shijie Xu and Akira Onoda*. "Accurate and Rapid Prediction of Protein pK_a: Protein Language Models Reveal the Sequence-pK_a Relationship." *Journal of Chemical Theory and Computation* 21(7): 3752–3764, **2025**, *DOI:* https://doi.org/10.1021/acs.jctc.4c01288
4. Shijie Xu and Akira Onoda*. "PsiPartition: Improved Site Partitioning for Genomic Data by Parameterized Sorting Indices and Bayesian Optimization." *Journal of Molecular Evolution* 92: 874–890, **2024**, *DOI:* https://doi.org/10.1007/s00239-024-10215-7
5. Shijie Xu and Akira Onoda*. "Accurate and Fast Prediction of Intrinsically Disordered Protein by Multiple Protein Language Models and Ensemble Learning." *Journal of Chemical Information and Modeling* 64(7): 2901–2911, **2024**, *DOI:* https://doi.org/10.1021/acs.jcim.3c01202

Presentations

- Shijie Xu and Akira Onoda, "Accurate and Fast Prediction of Metal-binding Sites in Proteins Using Deep Learning Methods", 52nd Biomolecule Science Symposium, 14th May - 16th May 2026, Himeji, Japan. [Oral]
- Shijie Xu and Akira Onoda, "Accurate and Fast Prediction of Protein pK_a by A Protein Language Model", the 19th Symposium on Biorelevant Chemistry, 2nd Sep - 4th Sep 2025, Kyoto, Japan. [Poster]
- Shijie Xu and Akira Onoda, "Accurate and Fast Prediction of Protein pK_a by Protein Language Model", 51st Biomolecule Science Symposium, 26th Jun - 27th Jun 2025, Sapporo, Japan. [Oral]
- Shijie Xu and Akira Onoda, "Accurate and Fast Prediction of Protein pK_a by Protein Language Model", 8th ICReDD International Symposium, 22nd Oct - 24th Oct 2024, Sapporo, Japan. [Poster]
- Shijie Xu and Akira Onoda, "Accurate Prediction of Intrinsically Disordered Protein by Protein Language Models", the 24th Annual Meeting of the Protein Science Society of Japan, 11th Jul - 13th Jul 2024, Sapporo, Japan. [Poster]
- Shijie Xu and Akira Onoda, "Accurate Prediction of Protein pK_a by Geometric Deep Learning", the 104th CSJ annual meeting, 18th Mar - 21st Mar 2024, Funabashi, Japan. [Oral]
- Shijie Xu and Akira Onoda, "Accurate Prediction of Intrinsically Disordered Protein by Protein Language Models", the 17th Biorelevant Chemistry Symposium, 8th Sep - 10th Sep 2023, Chiba, Japan. [Poster]
- Shijie Xu and Akira Onoda, "Fast and Accurate Prediction of Intrinsically Disordered Protein by Protein Language Model", the 103rd CSJ annual meeting, 22nd Mar - 25th Mar 2023, Chiba, Japan. [Oral]

Awards, Fellowships, and Grants

- Program for Supporting Challenging and Interdisciplinary Research Field, 2022, 2023, 2024
- Hokkaido University DX Doctoral Fellowship, 2022-2025
- Recipient of the Outstanding Student Scholarship, University of Science and Technology of China, 2016

- S.-T. Yau College Student Mathematics Contest, Beijing, China, 2016
- Recipient of the Outstanding Freshman Scholarship, University of Science and Technology of China, 2014
- Winner of the First Prize in the National High School Mathematics League (province level), Hefei, Anhui, China, 2012